

Autonomous Ops Analytics Report

LLM-Orchestrated Pipeline (Python execution engine)

2026-09-10

Run ID: `sample_run`

1 Executive Summary

Automated analysis of 200 rows x 6 columns produced 4 key finding(s). Strongest linear association is `temperature_rollmean3` vs `temperature_lag1` ($r=0.863$).

2 Dataset Profile

Rows: 200, Columns: 6. Timestamp: `timestamp`.

- `timestamp` (str) — missing 0.0%
- `temperature` (float64) — missing 0.5%
- `pressure` (float64) — missing 0.0%
- `vibration` (float64) — missing 0.0%
- `status` (str) — missing 0.5%
- `fault` (int64) — missing 0.0%

3 Analysis Plan (Stage-1)

- cleaning: [`"drop_duplicates"`, `"median_mode_fill"`]
- analyses: [`"correlation"`, `"anomaly_detection"`, `"forecasting"`]
- feature_engineering: [`"time_features"`, `"lags"`, `"rolling_stats"`]
- visualizations: [`"histograms"`, `"time_series"`, `"forecast_plot"`]
- supervised: `"random_forest_classifier"` target=`fault`

4 Computed Results

4.1 Correlation

```
{  
  "status": "ok",  
  "columns": [  
    "temperature",
```

```

"pressure",
"vibration",
"fault",
"hour",
"dayofweek",
"month",
"temperature_lag1",
"pressure_lag1",
"vibration_lag1",
"temperature_rollmean3",
"pressure_rollmean3",
"vibration_rollmean3"
],
"top_pairs": [
{
  "a": "temperature_rollmean3",
  "b": "temperature_lag1",
  "abs_corr": 0.863
},
{
  "a": "temperature_lag1",
  "b": "temperature_rollmean3",
  "abs_corr": 0.863
},
{
  "a": "temperature",
  "b": "temperature_rollmean3",
  "abs_corr": 0.839
},
{
  "a": "temperature_rollmean3",
  "b": "temperature",
  "abs_corr": 0.839
},
{
  "a": "pressure_lag1",
  "b": "pressure_rollmean3",
  "abs_corr": 0.714
},
{
  "a": "pressure_rollmean3",
  "b": "pressure_lag1",
  "abs_corr": 0.714
},
{
  "a": "temperature_rollmean3",
  "b": "pressure_rollmean3",
  "abs_corr": 0.668
},
{
  "a": "pressure_rollmean3",
  "b": "temperature_rollmean3",
  "abs_corr": 0.668
}

```

```

    },
    {
      "a": "pressure_rollmean3",
      "b": "pressure",
      "abs_corr": 0.665
    },
    {
      "a": "pressure",
      "b": "pressure_rollmean3",
      "abs_corr": 0.665
    }
  ]
}

```

4.2 Clustering

```
{}
```

4.3 Anomaly

```

{
  "status": "ok",
  "method": "IsolationForest",
  "n_anomalies": 10,
  "anomaly_pct": 5.0
}

```

4.4 Forecast

```

{
  "status": "ok",
  "method": "naive-last-value-backtest",
  "target": "temperature",
  "mae": 1.9407,
  "rmse": 2.2605,
  "n_train": 160,
  "n_test": 40
}

```

4.5 Supervised

```

{
  "status": "ok",
  "task": "classification",
  "method": "random_forest_classifier",
  "accuracy": 1.0,
  "f1_macro": 1.0,
  "saved_pipeline": "models/supervised_pipeline.joblib"
}

```

5 Key Findings (Stage-2)

- Strongest linear association is temperature_rollmean3 vs temperature_lag1 ($-r=0.863$).

- Anomaly screening flagged 10 rows (5.0%) via IsolationForest.
- Naive backtest on 'temperature': RMSE=2.2605, MAE=1.9407.
- Supervised classification accuracy=1.0, F1-macro=1.0.

6 Recommendations

- Validate top associations with domain context before acting.
- Review flagged anomalies for data-quality vs operational causes.
- Promote the saved pipeline only after holdout validation.

7 Limitations

- LLM narrative is evidence-grounded but not a substitute for domain review.

8 Figures

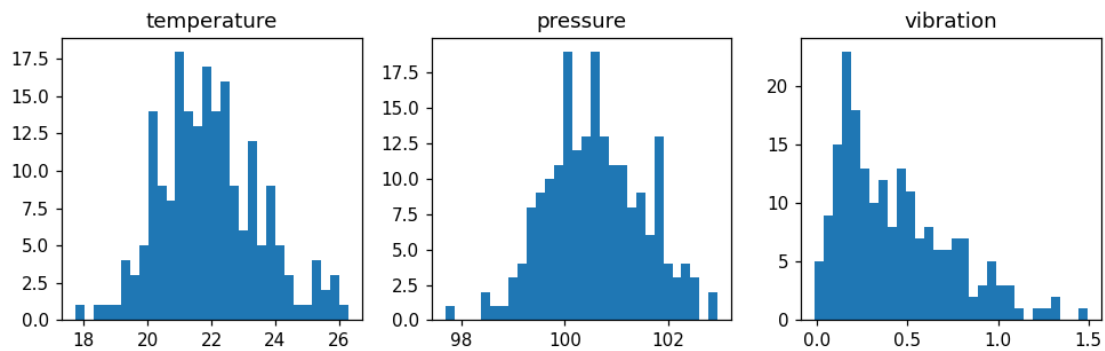


Figure 1: histograms.png

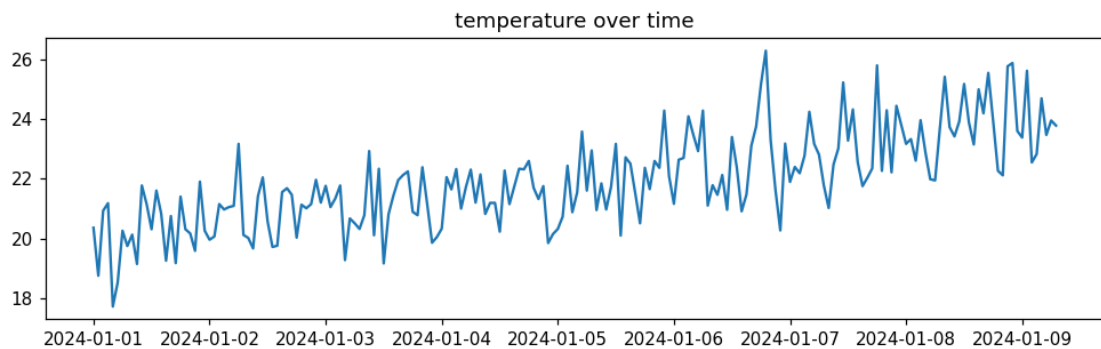


Figure 2: time_series.png

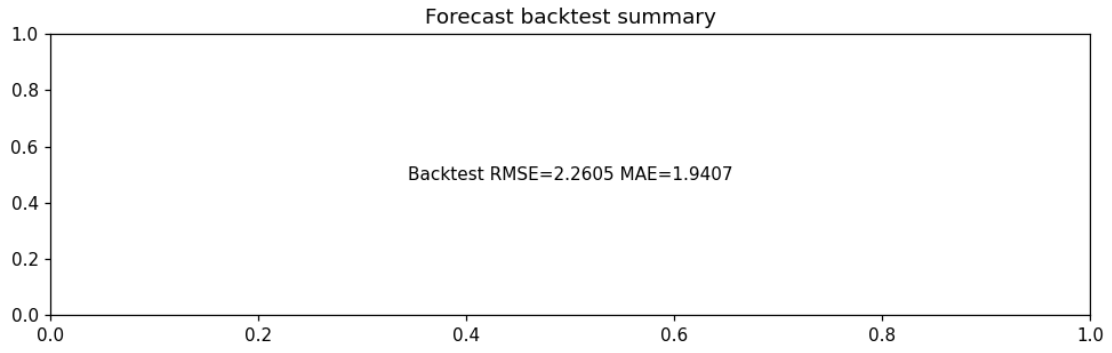


Figure 3: forecast_plot.png

9 Reproducibility

All numbers in this report were computed locally by the deterministic Python engine; the LLM planned the methodology and interpreted the evidence. See `profile.json`, `gemini_plan.json`, `manifest.json`.